

Python Developer Internship Assignment

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1. Project Overview

A Django 6.x web application built with two apps — users and products — each connected to its own SQLite database via a custom database router. The project demonstrates custom user authentication, post creation with `@login_required`, a ForeignKey without database constraint, admin customization, and public product listing.

2. Tech Stack

- Backend: Python 3, Django 6.x
- Frontend: Bootstrap 5 (CDN)
- Database: SQLite (two separate files)
- Configuration: python-decouple (environment variables)
- Version Control: Git & GitHub

3. Folder Structure

```
tradexa_assignment/
├── config/
│   ├── settings.py
│   ├── urls.py
│   ├── routers.py
│   └── ...
├── users/
│   ├── models.py
│   ├── views.py
│   ├── forms.py
│   ├── admin.py
│   └── templates/users/...
├── products/
│   ├── models.py
│   ├── views.py
│   ├── admin.py
│   └── templates/products/...
├── users_db.sqlite3
├── products_db.sqlite3
├── manage.py
├── requirements.txt
├── .env.example
└── README.md
```

4. Models

User (`users.models.User`)

- Inherits from `AbstractUser`
- Provides `username`, `first_name`, `last_name`, `email`, `password`

Post (`users.models.Post`)

- `user`: `ForeignKey` to `User` with `db_constraint=False`, `on_delete=CASCADE`
- `text`: `TextField`
- `created_at`, `updated_at`: auto-managed `DateTime` fields

Product (`products.models.Product`)

- `name`: `CharField`
- `weight`: `DecimalField` (kg)
- `price`: `DecimalField`
- `created_at`, `updated_at` — both displayed on the public `Products` page to show when a product was last modified

5. Database Router Explanation

- Router file: config/routers.py
- Routes users app to default database (users_db.sqlite3)
- Routes products app to products_db database (products_db.sqlite3)
- Prevents cross-database relations via allow_relation()
- Ensures migrations run only on the correct database via allow_migrate()
- Every read/write operation is directed based on app_label

6. Authentication Flow

- Custom user model set via `AUTH_USER_MODEL = 'users.User'`
- Django's built-in `LoginView` and `LogoutView` used
- Post creation view decorated with `@login_required` — unauthenticated users are redirected to login with `?next=/create/`
- Session-based authentication handled by Django middleware
- Navbar shows dynamic links based on `user.is_authenticated`
- Edit post view (`@login_required`) allows only the post owner to edit their own posts — accessible via `/edit/<id>/`

7. Admin Customization

- All models registered with custom ModelAdmin classes
- UserAdmin inherited for user management
- PostAdmin and ProductAdmin include list_display, list_filter, search_fields
- Admin accessible only to staff users

8. Screenshots

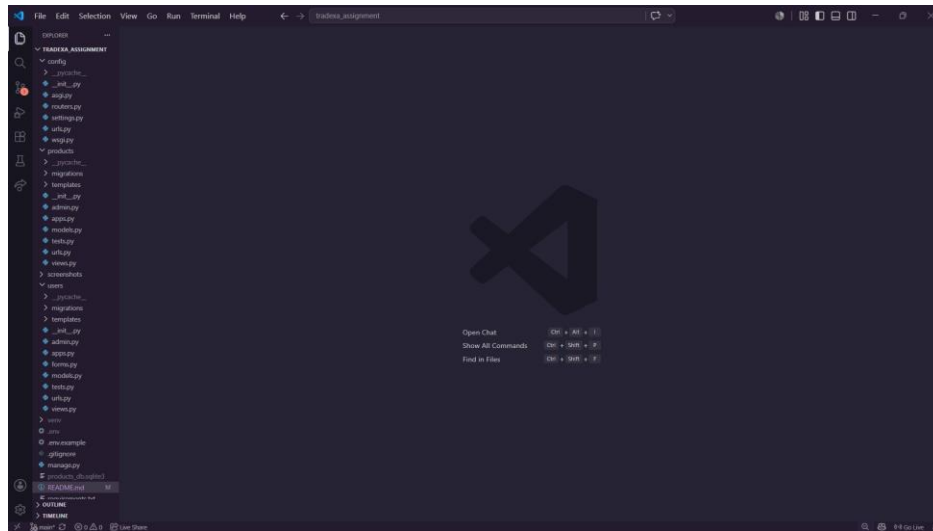


Figure 1: Project Folder Structure

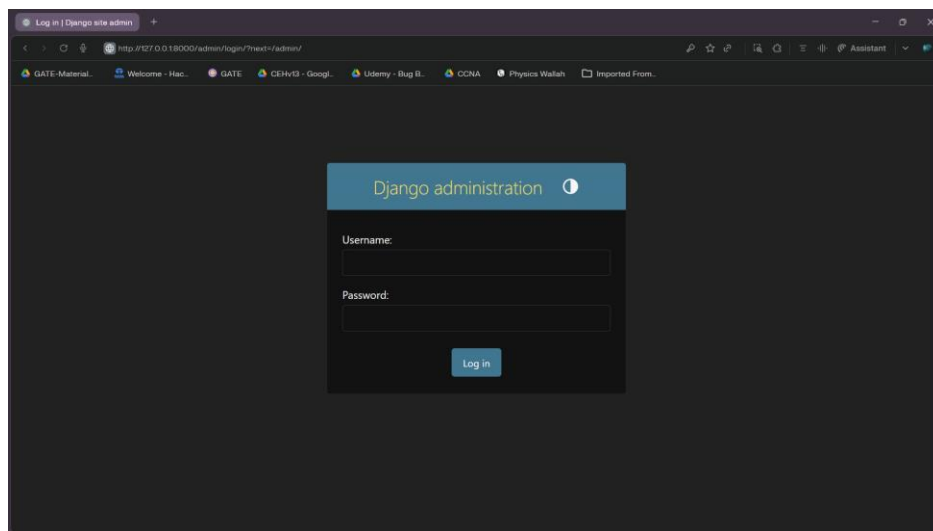


Figure 2: Admin Login Page

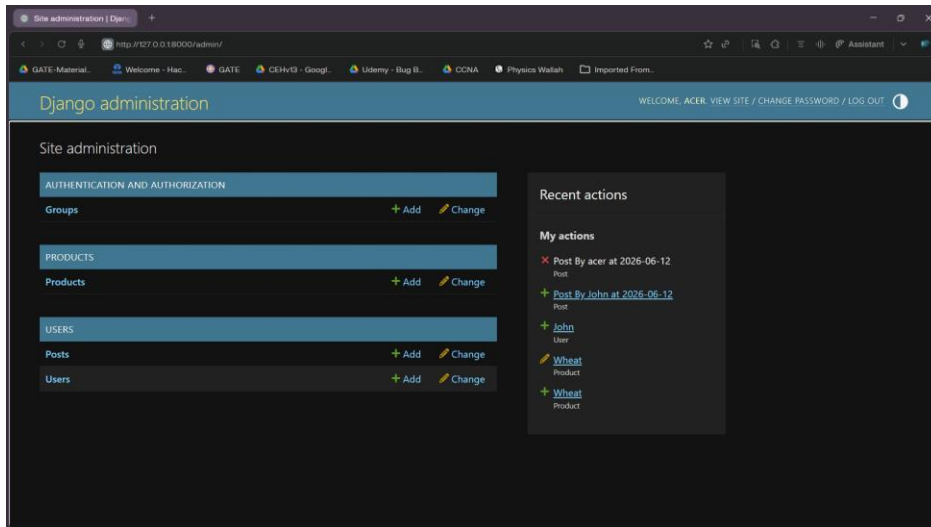


Figure 3: Admin Dashboard

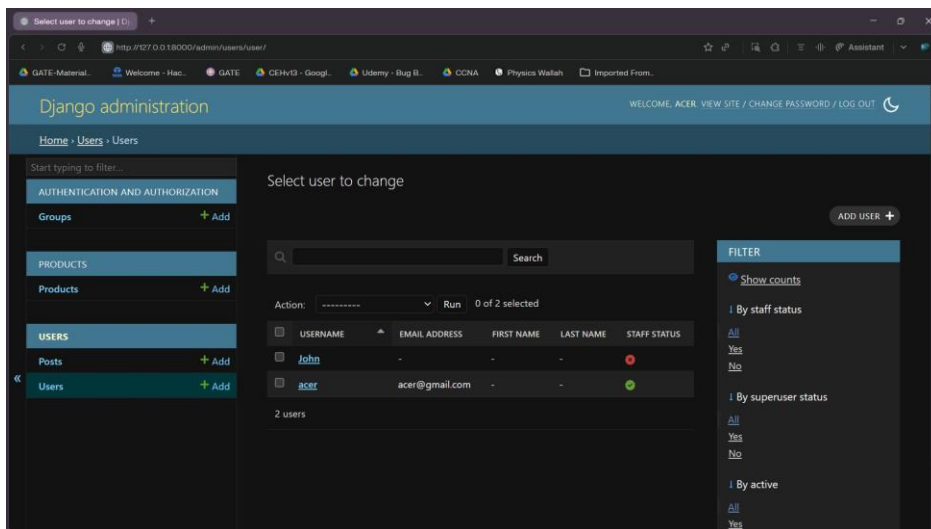


Figure 4: Users List (Admin)

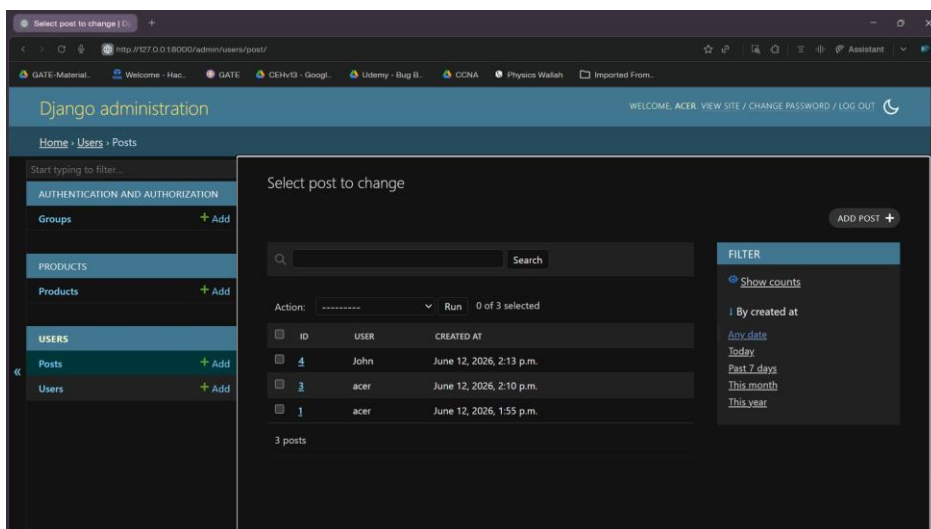


Figure 5: Posts List (Admin)

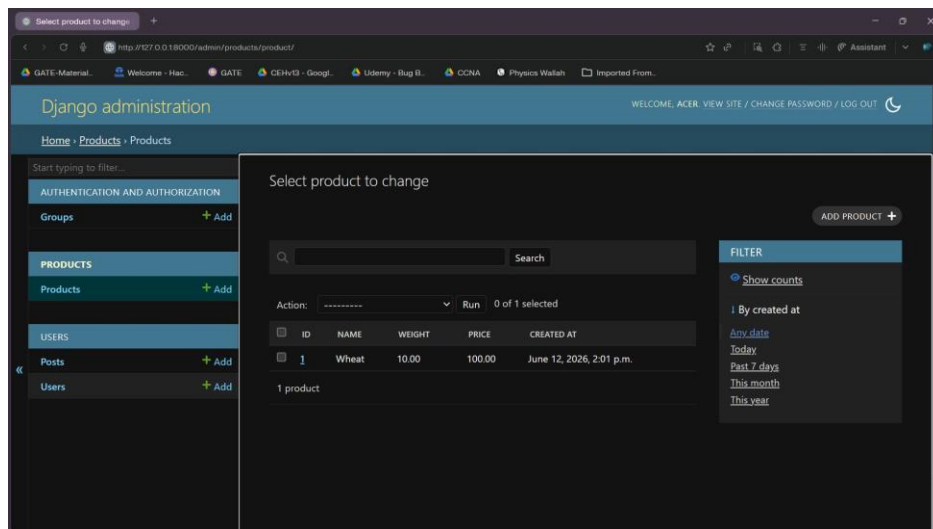


Figure 6: Products List (Admin)

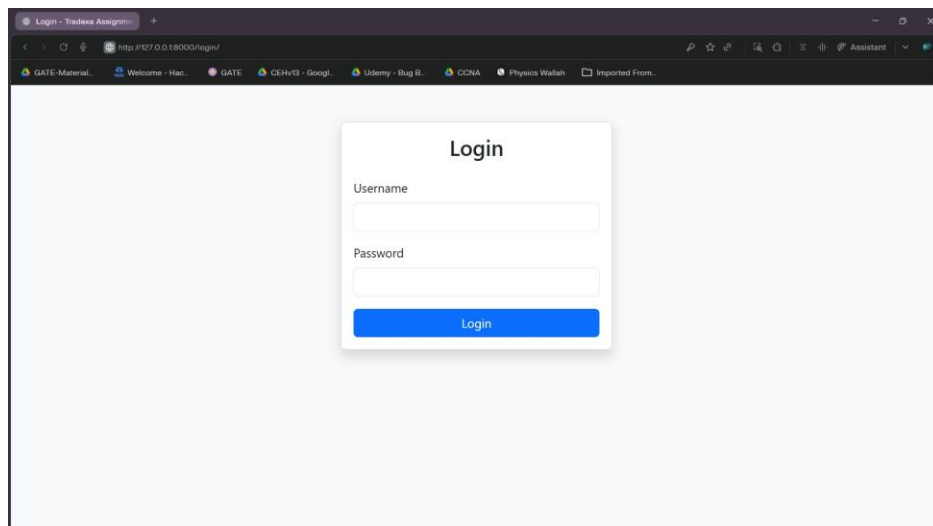


Figure 7: User Login Page

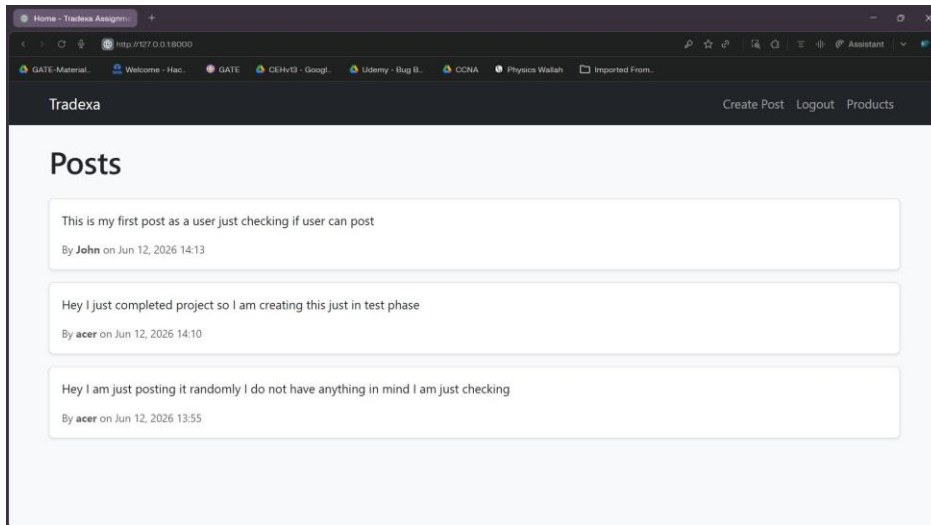


Figure 8: Home Page (Posts Feed)

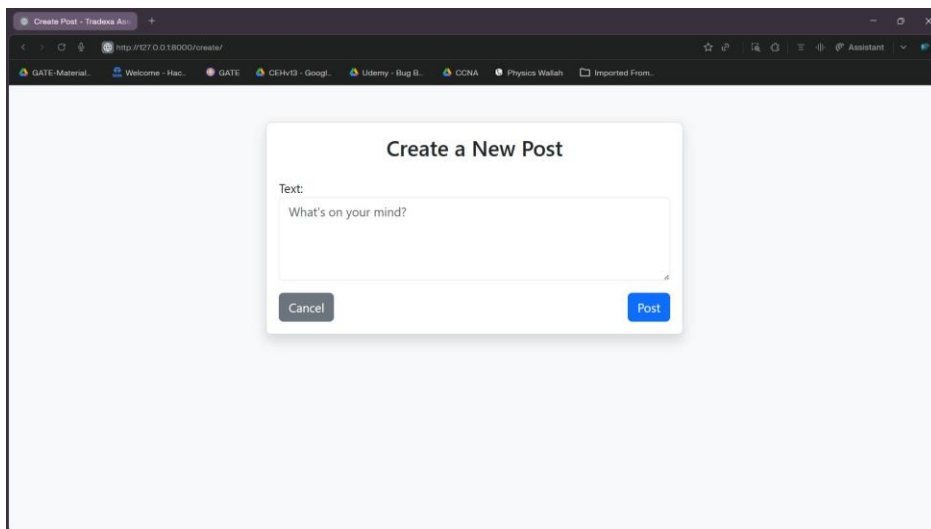


Figure 9: Create Post Page

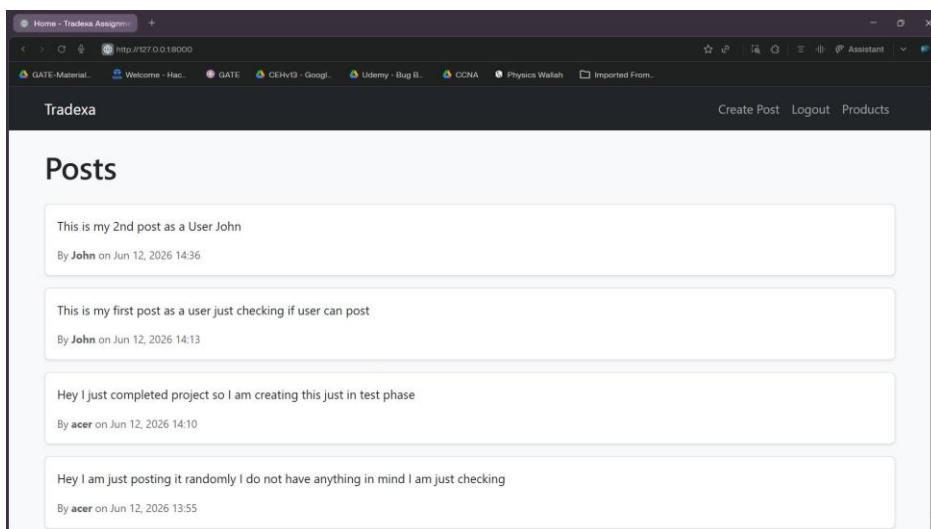
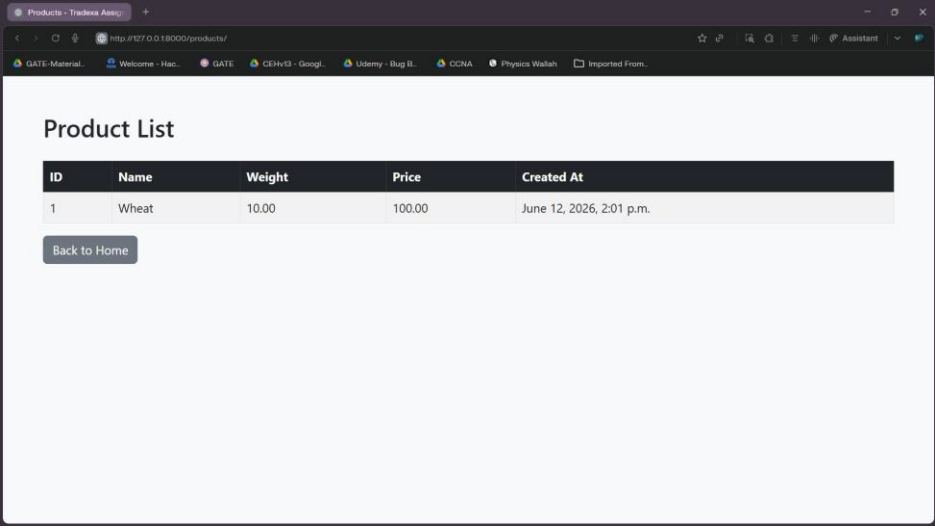


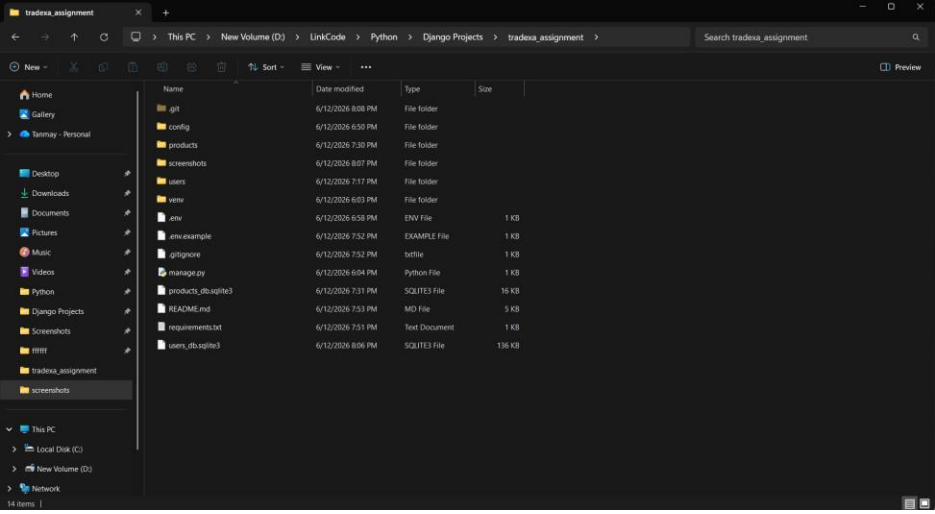
Figure 10: Successful Post Created



ID	Name	Weight	Price	Created At
1	Wheat	10.00	100.00	June 12, 2026, 2:01 p.m.

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Figure 11: Public Products Page



Name	Date modified	Type	Size
api	6/12/2026 8:08 PM	File folder	
config	6/12/2026 6:50 PM	File folder	
products	6/12/2026 7:30 PM	File folder	
screenshots	6/12/2026 8:07 PM	File folder	
users	6/12/2026 7:17 PM	File folder	
venv	6/12/2026 6:03 PM	File folder	
db	6/12/2026 6:58 PM	ENV File	1 KB
example	6/12/2026 7:52 PM	EXAMPLE File	1 KB
apiignore	6/12/2026 7:52 PM	textfile	1 KB
manage.py	6/12/2026 6:04 PM	Python File	1 KB
products_db.sqlite3	6/12/2026 7:31 PM	SQLite3 File	16 KB
README.md	6/12/2026 7:51 PM	MD File	5 KB
requirements.txt	6/12/2026 7:51 PM	Text Document	1 KB
users_db.sqlite3	6/12/2026 8:06 PM	SQLite3 File	136 KB

Figure 12: Database Files

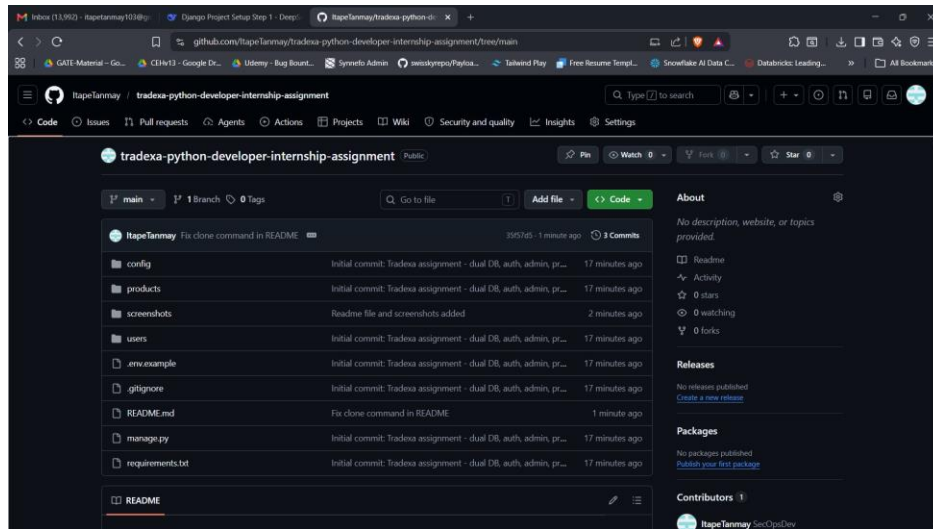


Figure 13: GitHub Repository

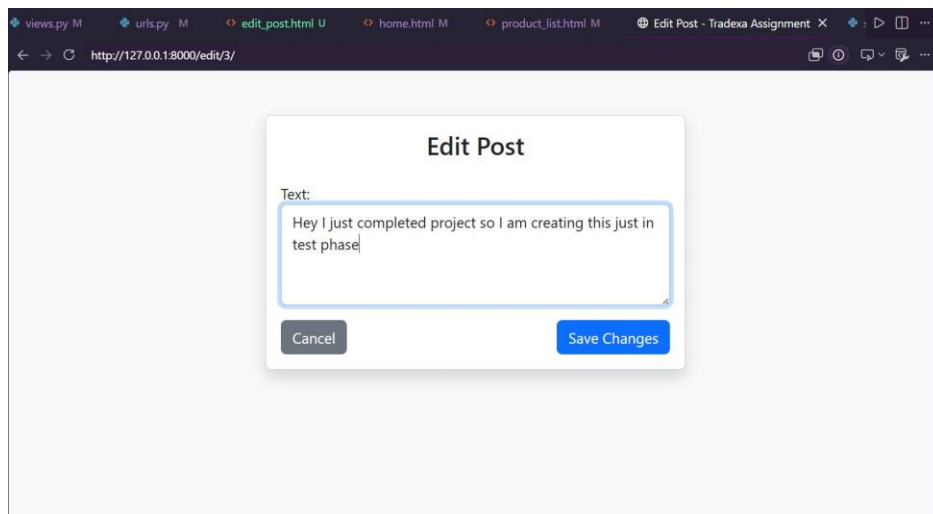


Figure 14: Edit Post Page

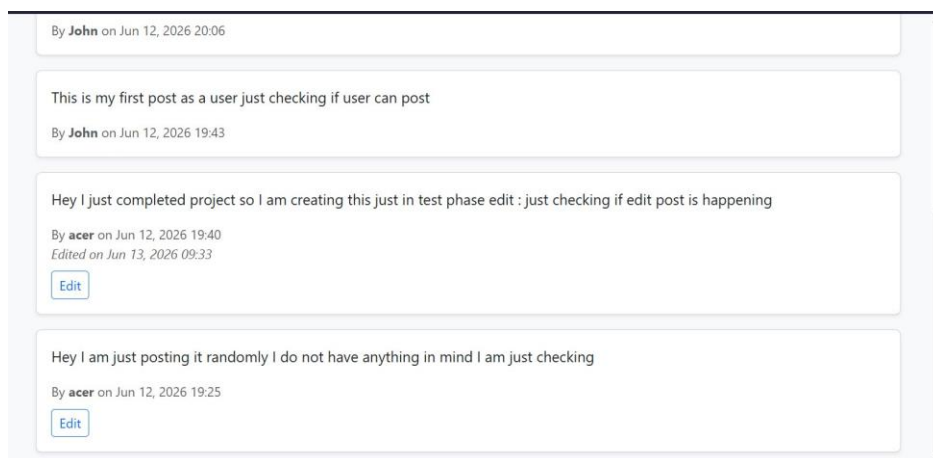


Figure 15: Edit Post – Success (Updated Feed)

Product List

ID	Name	Weight	Price	Created At	Updated At
1	Wheat	10.00	100.00	Jun 12, 2026 19:31	Jun 12, 2026 19:31

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Figure 16: Products Page with Updated At Column

9. GitHub Link

<https://github.com/ltapeTanmay/tradexa-python-developer-internship-assignment>

10. Conclusion

This project demonstrates the ability to build a multi-app Django application with advanced configurations like dual databases, database routing, custom authentication, and admin customization. All assignment requirements have been fulfilled, and the code follows industry best practices (PEP 8, separation of concerns, secure config via environment variables).